

Probability of a set of observed words $w_t \in V$:

$$\prod_k p_k^{n_k}$$

To make things more tractable, take the log probability

$$\sum_k n_k \log p_k$$

maximize it with constraint $\sum_k p_k = 1$, use method of Lagrange multipliers to set up the Lagrangian L :

$$L = \sum_k n_k \log p_k + \lambda(1 - \sum_k p_k)$$

Differentiating L with respect to p_k and setting it to zero gives:

$$\frac{\partial L}{\partial p_k} = \frac{n_k}{p_k} - \lambda = 0 \Rightarrow p_k = \frac{n_k}{\lambda}$$

Summing both sides, and using the constraint $\sum_k p_k = 1$:

$$\sum_k p_k = \sum_k \frac{n_k}{\lambda} = 1 \Rightarrow \lambda = \sum_k n_k$$

Plugging in this value for λ :

$$p_k = \frac{n_k}{\lambda} = \frac{n_k}{\sum_k n_k}$$

this p is optimal